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MILITARY SPECIFICATION

FILTER ASSEMBLY, FLUID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This document covers filter assemblies, fluid used for the removal of insoluble impurities from power plant turbine oil and diesel engine lubricating oil (additive type).

1.2 Classification.

1.2.1 Types, classes, styles and applications. The filter assemblies covered by this document shall be of the following types, styles, classes, and applications as specified (see 6.2).

Type I - Single filter, assembly
Type II - Dual filter, assemblies

Style A - Full flow operation
Style B - Bypass operation

Class 1 - Without heater
Class 2 - With electric heater
Class 3 - With steam heater
Class 4 - With hot water heater

Application A - Turbine Oil
Application B - Diesel Engine Lubricating Oil

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4330

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1.2.2 Part number designation. Filter assemblies covered by this document shall be identified by a part or identifying number (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

F-F-351 - Filter and Filter Elements, Fluid, Pressure, Lubricating Oil, Bypass and Full Flow.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-P-40 - Packaging and Packing of Hand Tools.

MILITARY

MIL-V-3 - Valves, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of.
MIL-P-116 - Preservation, Methods of.
MIL-V-173 - Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-P-514 - Plate, Identification, Instruction and Marking, Blank.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Provisioned Items (Repair Parts); Packaging of.
MIL-L-27502 - Lubricating Oil, Aircraft Turbine Engine, Ester Base.

STANDARDS

FEDERAL

FED-STD-595 - Color.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking And Waterproofing, With Appropriate Test Methods.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

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(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B31.1 - Power Piping.
ANSI B40.1 - Gauges, Pressure Indicating Dial Type, Elastic Element
ANSI B93.31 - Multi-pass Method for Evaluating the Filtration Performance of a Fine Hydraulic Fluid Power Filter Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN PETROLEUM INSTITUTE. (API)

API-STD-614 - Lubrication, Shaft-Sealing, and Control-Oil Systems for Special-Purpose Applications.

(Application for copies should be addressed to the American Petroleum Institute, 1220 L. Street NW, Washington, DC 20005.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS. (ASME)

ASME Boiler and Pressure Vessel Code Section VIII, Division I - Pressure Vessels.

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION. (NFPA)

NFPA 70 - National Electric Code.

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The filter assembly shall include shell, cover, removable filter elements, gages, piping, valves, and additional equipment required for the satisfactory operation of the unit. The equipment shall be complete with all accessories necessary for operation when connected to the oil lines, drain, and to the applicable heat source.

3.2 First article. When specified (see 6.2), the contractor shall furnish one filter assembly of each classification included in the contract for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Standard Commercial Product. The filter assembly shall, as a minimum, be in accordance with the requirements of this document. Additional or better features, which are not specifically prohibited by this document but which are a part of the manufacturer's standard commercial product, shall be included in the filter assembly being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs or brochures and represents the latest production model.

3.4 Materials. Materials shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this document are to be new; however, recovered raw materials shall be used to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials reprocessed to become a source of raw materials, as opposed to virgin raw materials which include previously unused metal or metal ore, any undeveloped resource, or new technology which will become a source of raw material. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed, unless otherwise specified.

3.5 Interchangeability. All filter assemblies of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

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3.6 Design and construction. The filter assemblies shall be designed to withstand strains, jars, vibrations, and other conditions incidental to shipping, storage, installation, and operation and to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to equipment. The filter assemblies shall be furnished complete in all details including those not specifically mentioned herein.

3.6.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

3.6.2 Capacity.

3.6.2.1 Application A. The design rate of flow in gallons per hour (gph) at 100 degrees (^{o}) Fahrenheit (F) ($\pm$ 4°F) through the filter element shall not be less than that specified (see 6.2).

3.6.2.2 Application B. The design rate of flow of oil in gph at 200 Seconds Saybolt Universal (SSU) viscosity through the filter element shall be not less than that specified (see 6.2).

3.6.3 Components. Type I filter assembly shall consist of one shell and associated parts. Type II filter assembly shall consist of two shells and associated parts arranged for parallel operation.

3.6.3.1 Shell. The shell of the filter element container shall be of pressed steel or welded steel construction and shall enclose, support, and protect all elements. Separate compartments shall be provided for dirty oil and filtered oil. Supports shall be provided for base or side mountings adequate to support the unit in service. Tie-bolts, springs, or equivalent means shall hold the filter elements in recesses or other devices for receiving and loading the elements. Unless a greater working pressure is specified (see 6.2), shell shall be designed and constructed for 100 pound-force per square inch gage (psig) working pressure. All filter shells shall conform to ASME Boiler and Pressure Vessel Code, Section VIII, Division I.

3.6.3.2 Oil compartments. The compartment for dirty oil shall contain a sump for the collection of rejected particles. The inlet oil opening shall be above the sump. A drain connection shall lead from the sump. If the outlet oil connection is so placed that a space below the entrance level of the outlet could serve as a sump, the drain connection shall be in the bottom of this space. The shell shall be so designed and constructed that dirt dropping from filter elements is prevented from entering the clean oil passages. The inlet oil passages shall be such that, over the inlet area of each filtering element, oil shall be uniformly distributed under all conditions of temperature, viscosity, and oil contaminant content. All joints between compartments shall be such that oil flows only through the designed connections. All heated filter assemblies shall be insulated. Filter insulation shall be protected from contamination and damage by a removable steel metal jacket. Filters heated by immersion heaters or pipes and steam or hot fluid jacketed models shall have

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insulation in contact with hot exterior walls. Filters heated by hot pipes or electric heating elements in contact with the filter wall shall have the insulation placed between the pipes or heating elements and the removable outer jacket.

3.6.3.3 Cover. The removable cover shall be quick opening, constructed of steel, and secured to the shell by nuts, wing nuts, or toggle bolts. The cover shall provide access to all components and not cause a break in connections. Fittings shall not be attached to the cover. Lifting devices with integral supports shall be provided for covers weighing over 30 pounds (lb).

3.6.3.4 Filter elements (cartridges). Each filter assembly shall employ a sufficient number of elements connected in the manner necessary to achieve the required performance (see 3.7). The filter elements shall be of the following:

a. Repackable or throwaway type (see 6.2).
b. Material shall be cellulose, cloth, or other substance (see 6.2).
c. Conform to type IV, class 3 or 4, of F-F-351 (see 6.2).
d. Listed or be approved for listing in the applicable Qualified Products List.

3.6.3.4.1 Application A. The pressure drop for clean filter elements or cartridges shall not exceed 15 percent of the total allowable dirty pressure (terminal pressure) drop nor shall it exceed 5 pound force per square inch (psi) at 100°F operating temperature and normal flow. Elements or cartridges shall have a minimum collapsing differential pressure of 70 psi. The elements or cartridges shall pass the applicable qualification tests specified in ANSI B93.31.

3.6.3.4.2 Application B. The pressure drop for clean filter elements or cartridges shall not exceed 7 psi of clean test oil at the engine design flow rate. The elements or cartridges shall pass the applicable qualification tests specified in F-F-351.

3.6.3.5 Filter element retainer. Filter elements shall be secured in place either by a nut or wing nut over a metallic washer with a soft gasket between the washer and the top of the core or case or by a spring that imposes a load not less than 100 lb or more than 500 lb. Support plates or recesses in the shell shall be used to locate the lower end of the element. Gaskets shall be used to seal the lower end of the elements. Dirty oil shall not bypass the filter element.

3.6.3.6 Gaskets. The gasket shall be used between the cover and the shell. Gaskets shall be of a material that shall not shrink, stretch, become porous or gummy, or disintegrate in service when subjected to lubricating oil and contaminants at temperatures ranging from 0° to 250° F.

3.6.4 Gages. Gages shall be mounted on the filter shell and shall conform to ANSI B40.1, 4 1/2 inch nominal dial diameter, 0 - 100 psi range. A valve and fittings shall be provided for mounting each gage.

3.6.5 Shafts and rods. Operating shafts, rods, and valve stems shall be of corrosion resistant metal and shall be effectively sealed against leakage.

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3.6.6 Seals. Seals for shafts, rods, and valve stems shall be so designed as not to react chemically with the lubricating oil or stick to the shaft, rod, or stem. Packings shall be easily adjustable and shall be removable only by disassembly of the packing retainer. Seals of the solid bushing type shall be made of materials designed to operate without scoring or galling.

3.6.7 Piping. Bypass piping or passages, incorporating a pressure relief valve set to open at the maximum designed pressure drop over the element, shall be provided. The bypass shall be so designed that the filter pressure is not in excess of 45 psig at a rate not less than 125 percent of the specified flow. Oil inlet and oil discharge pipes shall be provided with threaded connections. All connections shall be arranged as to make possible removal of filter elements without breaking any outside connections. Type II filter assemblies shall be provided with ferrous valves so arranged that either filter assembly may be removed from service without disturbing operation of the other filter. Pipe and fittings shall be ferrous and shall conform to ANSI B31.1. Pipe threads shall be National Pipe Taper threads. The nominal size of oil inlet and discharge pipes shall be not less than the following:

TABLE I. Nominal pipe size

Gph	Nominal pipe size
0 - 345	1/2 in (12.7 mm)
346 - 765	3/4 in (19.05 mm)
766 - 980	1 in (25.4 mm)
981 - 1700	1 1/4 in (31.75 mm)
1701 - 2300	1 1/2 in (38.1 mm)
2301 - 3800	2 in (50.8 mm)

Inlet and discharge connections for larger flows shall be in accordance with the manufacturer's standard practice.

3.7 Performance. The filter assembly shall be designed to remove at least 98 percent of particles equal to or larger than the minimum size as specified herein. When clean oil is used, the pressure drop through a clean filter element shall not exceed the requirements of 3.6.3.4. Not more than 10 percent of any oil additive shall be removed during continuous use of the filter. The type, brand, and grade of lubricating oil to be filtered shall be as specified (see 6.2)

3.7.1 Solids retaining capacity.

3.7.1.1 Application A. Each filter element shall have not less than the minimum AC Fine Test Dust (ACFTD) capacity (a) specified (see 6.2). At the terminal pressure drop, the filtration ratio shall not be greater than the maximum filtration ratio specified (see 6.2).

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3.7.1.2 Application B. Unless a greater holding capacity is specified (see 6.2), each filter element shall be capable of retaining 1.1 lb of the solid test contaminant of F-F-351. The pressure drop across the filter after the required weight of test contaminant has been retained shall not exceed 40 psig.

3.8 Control valves. Control valves and regulators shall have valve seating surfaces of corrosion resisting metal and shall be designed to operate without pitting, sticking, or galling. Valve bodies shall conform to ANSI B31.1.

3.9 Heater. When a heated filter assembly is required, the heater shall be capable of raising the oil temperature 50°F for the specified flow of oil (see 3.6.2 and 6.2). The filter assembly design shall be such that no area in contact with the oil shall be heated to a temperature higher than 300°F. An insulation jacket shall be furnished.

3.9.1 Electric Heating. Electric heaters shall be of the resistance type with the electrical characteristics as specified (see 6.2). The heaters shall be properly grounded and present no electric hazards to personnel or equipment.

3.9.1.1 Temperature control, magnetic contactor, and start and stop switch. A temperature control, provided for operation with a magnetic contactor, shall function to maintain the incoming oil reaching the filter element at temperatures between 170°F and 250°F, shall be adjustable over a minimum range of 60°F, and shall maintain the set temperature within ±8°F. The magnetic contactor, which shall make and break the heating element circuit, shall have contact points designed for continuous operation at maximum rated load. The temperature controller and associated contactor shall be housed in a totally enclosed case mounted on the shell at a location that neither interferes with removal of filter elements nor exposes the unit to damage as a result of the filter removal process. The contactor may be housed separately from the temperature controller. A totally enclosed, manually operated, start and stop switch to activate the heater shall be provided. The preferred mounting location for all electrical components is on the filter shell. When the heating device is too large to permit mounting on the filter shell, the device shall be mounted on a separate panel.

3.9.1.2 Electrical wiring and components. All electrical wiring and components shall be designed for the power specified, and have sufficient capacity for uninterrupted service while carrying the maximum rated load. Electrical components shall be properly connected together by suitable wiring and accessories. All components and wiring connections shall be approved by the Underwriters Laboratories for the intended application and installed in accordance with applicable requirements of NFPA 70.

3.9.1.3 Fungus resistance. When specified (see 6.2), electrical components and circuit elements, including terminal and circuit connections, shall be coated with varnish conforming to MIL-V-173, except that:

a. Components and elements inherently inert to fungi or in hermetically sealed enclosures need not be coated.
b. Current carrying contact surfaces, such as relay contact points, shall not be coated.

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3.9.2 Steam and hot water heating. Steel coils, passages, or chambers shall be designed for maximum steam temperature of 300°F, and a maximum steam pressure of 50 psig. Heaters shall meet the temperature control requirements with the steam or hot water pressure and temperature specified.

3.9.2.1 Temperature control. A temperature control shall function to maintain the incoming oil reaching the filter element at temperatures between 170°F and 250°F, shall be adjustable over a minimum range of 60°F, and shall maintain the set temperature within ±8°F. The control shall include a control or regulator valve suitable for regulating the flow of steam or hot water and shall have a regulating mechanism designed for a year's continuous operation at maximum rated load. The temperature controller and associated components shall be located such that it neither interferes with removal of filter elements nor exposes the unit to damage as a result of the filter removal process. The preferred mounting location for all components is on the filter shell. When the heating equipment is too large to permit mounting on the filter shell, the equipment shall be mounted on a separate panel.

3.10 Codes and standards. The filter assembly shall conform to the requirements of ANSI B31.1, ASME Boiler and Pressure Vessel Code, Section VIII, Division I, and NFPA 70 as applicable.

3.10.1 Compliance. Prior to approval of the first shipment, the contractor shall submit to the contracting officer, or his authorized representative, satisfactory evidence that the filter assembly proposed to be furnished under this specification meets the above codes and standards.

3.10.2 Acceptable evidence. Acceptable evidence of meeting these requirements would be an ASME code stamp or a certified test report from a recognized independent testing laboratory indicating the filter assembly has been tested and conforms to the applicable standards. Such evidence must be acceptable to the contracting officer.

3.11 Instruction plates. Each filter assembly shall be furnished with suitably located, firmly attached instruction plates conforming to MIL-P-514, describing any special or important procedures for operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment. The following information shall be included in the instructions:

a_1. Capacity at 100°F operating temperature and normal flow for Application A filters.
a_2. Capacity at 200 SSU viscosity for Application B filters.
b. Pressure drop with clean filter element.
c. Filter element removal pressure.
d. Design shell operating pressure.
e. Power requirements for electric heaters, if applicable.
f. Design pressure and temperature for steam or hot water heaters, if applicable.

3.12 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat, conforming to FED-STD-595, shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they

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are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

3.13 Identification marking. Identification shall be permanently and legibly marked directly on the filter assembly or on a corrosion resisting metal plate securely attached to the filter assembly by the manufacturer. Identification shall include the manufacturer's model and serial number, name, and trademark to be readily identifiable to the manufacturer.

3.14 Workmanship.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.14.2 Bolted connections. Boreholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.14.4 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the casting's ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspections set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable reference documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. When a first article is required (see 3.2, 6.2, and 6.4), the inspection shall be performed on one filter assembly of each type, style, class and application included in the contract. This inspection shall include the examination of 4.4 and the tests of 4.5. The contracting officer will provide specific guidance on whether the first article must be a first article sample or may be a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract (see 6.2 and 6.4).

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the test of 4.5.1, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. When specified (see 6.2), sampling and inspection procedures shall be in accordance with MIL-STD-105. All filter assemblies of the same type, style, class, and application offered for delivery at one time shall be considered a lot for the purpose of inspection. The standard sample for first article inspection shall be ten percent of a lot with the minimum sample being not less than one unit. For each defective unit, two additional units shall be inspected until the lot has been rejected. If an inspection lot is rejected, the contractor may rework it to correct the defects or screen out the defective units and submit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection requirements. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots and shall be clearly identified as reinspected lots.

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4.4 Examination. Each filter assembly selected shall be examined for compliance with the requirements specified in section 3 of this document. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.5 Tests. Failure to pass any test shall constitute cause for rejection.

4.5.1 Hydrostatic test. Each filter assembly, without filter elements, shall be subjected to a hydrostatic test pressure of 150 psig in accordance with the applicable provisions of the ASME Boiler and Pressure Vessel Code. There shall be no distortion of the shell or leakage through the cover or from any weld or seam.

4.5.2 Pressure drop test. The filter assembly shall be installed complete with all accessories in a test facility suitably equipped to pump oil, measure the oil flow rate, the drop across the inlet and outlet connections, and, for heated filter assemblies, the temperatures of the influent oil, the effluent oil, and the ambient air. With new filter elements in place on the first article sample assembly, clean test oil shall be pumped through the assembly at the rate specified (see 3.6.2) while the discharge line is suitably throttled to yield the working pressure specified (see 3.6.3.1) at the inlet connection. Before the test, the filter assembly with elements installed shall be filled with test oil for a period of not less than 16 hours. For heated filter assemblies, the effluent oil shall be checked for a constant temperature of 250°F ±8°F.

4.5.2.1 Application A. The test oil shall be clean lubricating oil conforming to the applicable sections of MIL-L-27502. The temperature of the test oil shall be maintained at a temperature of 100°F. The pressure drop across the inlet and outlet connections shall not exceed 15 percent of the total allowable dirty pressure drop nor shall it exceed 5 psig.

4.5.2.2 Application B. The test oil shall be clean lubricating oil conforming to the applicable sections of F-F-351. The temperature of the test oil shall be maintained to yield a viscosity of 200 SSU, but in no case shall the temperature be above 150°F. The pressure drop across the inlet and outlet connections shall not exceed 7 psig.

4.5.3 Solids retaining capacity test. A single element of the first article sample (bypass or full flow as applicable) shall be tested for test material holding capacity at rated capacity and pressure.

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4.5.3.1 Application A. In accordance with ANSI B93.31, inject the test contaminant continuously from the agitated reservoir into the influent stream as close as practicable to the filter at a rate of 10 milligrams per liter of oil influent. The test contaminant shall be prepared in accordance with the applicable paragraphs of ANSI B93.31 and with the same test oil used in 4.5.2. The test run shall be continued until the pressure drop across the filter reaches the terminal pressure drop specified. The filter element ACFTD capacity and minimum filtration ratio shall not be less than specified (see 3.7.1).

4.5.3.2 Application B. In accordance with F-F-351, inject the test contaminant continuously from the agitated reservoir into the influent stream as close as practicable to the filter at a rate of 10 milligrams per liter of oil influent. The test contaminant shall be prepared in accordance with the applicable paragraphs of F-F-351 and with the same test oil used in 4.5.2. The test run shall be continued until the pressure drop across the filter reaches 40 psig. Each filter element shall be removed, allowed to drain for 1 hour and then weighed. The weight of the filter element shall be compared with the weight of a new identical element similarly drained for 1 hour after a 16 hour immersion into the test oil specified in 4.5.2. The weight of the dirty element shall be at least 1.1 lb heavier than the weight of the clean soaked new filter element and retain the solids specified (see 3.7.1).

4.6 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

5.1.1.2 Disassembly. Disassembly shall be minimum necessary to protect parts subject to damage or loss, and to accomplish reduction in cube. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.1.1.3 Cleaning and drying. Prior to application of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

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5.1.1.4 Matchmarking. Parts removed and mating parts on the equipment and attachments shall be matchmarked to facilitate reassembly. Parts and accessories removed and mating parts on the equipment shall be identified with weatherproof tags attached to mating parts and locations. Markings shall be applied to the tags with a waterproof material.

5.1.1.5 Unprotected surfaces. Unprotected exterior metal surfaces requiring the application of a contact preservative in accordance with MIL-P-116 and not specifically provided for herein shall be preserved with P-1 of MIL-P-116.

5.1.1.6 Valves, fittings, and piping. Valves, fittings, and piping, if removed from the equipment, shall be preserved and packaged in accordance with the level A requirements of MIL-V-3.

5.1.1.7 Gages. Gages subject to damage shall be protected as necessary to ensure the gage operates properly when the filter assembly is placed in service.

5.1.1.8 Filter elements. Filter elements shall be packaged method IC-3.

5.1.1.9 Electrical components. Magnetic contactors and electrical controls, if removed from the equipment, shall be preserved and packaged in accordance with the level A requirements of MIL-E-17555.

5.1.1.10 Technical Publications. Technical publications for each piece of equipment shall be preserved method IC-3.

5.1.1.11 Maintenance tools. Maintenance tools shall be preserved in accordance with level A preservation and packaging requirements of PPP-P-40.

5.1.1.12 Service Parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve service parts.

5.1.1.13 Consolidation. Disassembled items and technical publications for each filter assembly shall be consolidated in containers conforming to PPP-B-636, class weather-resistant. Contents shall be cushioned, blocked, and braced to prevent movement in accordance with MIL-STD-1186.

5.1.2 Commercial. The equipment shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or Commercial as specified (see 6.2).

5.2.1 Level A. Each complete filter assembly shall be packed in close-fitting boxes conforming to PPP-B-601, overseas type, or PPP-B-621, class 2. The contents shall be blocked, braced, and secured to prevent movement inside the container.

5.2.2 Level B. Each complete filter assembly shall be packed as specified for level A, except that boxes shall conform to PPP-B-601, domestic type; or PPP-B-621, class 1.

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5.2.3 Commercial. Packing shall be in accordance with ASTM D 3951.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The filter assemblies ordered using this specification are intended for cleaning and/or purifying additive turbine and diesel lubricating oils. A type I filter assembly is used when replacement of the filter element or maintenance operations allow cutting off the flow of oil to the filter assembly. A type II filter assembly is used when the flow of oil must be maintained during replacement of the filter elements or maintenance operations. Class 1 filter assemblies are used when maintenance of a particular viscosity is not essential for proper oil purification or environmental and operating conditions are such that viscosity of the lubricant is relatively constant. Heated filter assemblies are used when maintenance of a particular viscosity is essential for proper oil purification or environmental and operating conditions are such that viscosity of the lubricant widely varies. The equipment is not intended for nonadditive (straight mineral) type oils.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Type, style, class, and application required (see 1.2).

1. Class 2, 3, and 4 assemblies (see 3.9).

(a) Class 2

(1) Electrical requirements (see 3.9.1).
(2) When fungus and moisture resistance is required (see 3.9.1.3).

(b) Class 3 - Temperature and pressure (see 3.9.2).
(c) Class 4 - Temperature and pressure (see 3.9.2).

2. Application A and B assemblies

(a) Application A

(1) Design rate for flow of oil at 100°F (see 3.6.2).
(2) Minimum ACFTD capacity (see 3.7.1.1).
(3) Maximum filtration ratio (see 3.7.1.1).
(4) Terminal pressure drop (see 3.7.1.1).

(b) Application B

(1) Design rate for flow of oil at 200 SSU viscosity (see 3.6.2).
(2) Solids retaining capacity if other than 1.1 lb (see 3.7.1.2).

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(1) Class 1 - Without heater

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article filter assembly is required for inspection and approval (see 3.2, 4.2.1, and 6.4).

1. Location for first article inspection.
2. Notification requirements.
3. Special instructions.

e. Finish color shall be as specified (see 3.12).
f. Working pressure in psig for shells if other than 100 psig (see 3.6.3.1).
g. Type, class, and material of cartridge assembly (see 3.6.3.4).
h. Minimum particle size removal required (see 3.7).
i. Type, brand, and grade of lubricating oil (see 4.5.2 and 3.7).
j. Technical manuals required including contents and distribution (see 6.3.1).
k. When sampling in accordance with MIL-STD-105 is required (see 4.3).
l. Levels of preservation and packing required (see 5.1 and 5.2)

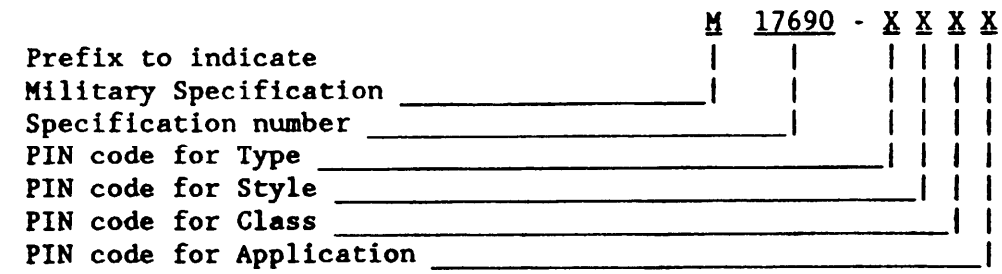
6.3 Data requirements. When this document is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL) incorporated into the contract. When provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.3.1 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals such as operator's manual including lubrication chart, parts manual, and/or maintenance, service, and repair manual are required, military specifications and standards that have been cleared and listed DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which must be acquired under separate contract line item in the contract.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results, and disposition of the first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Part or identifying number (PIN). The PIN to be used for filter assemblies acquired to this document are created as follows:



6.5.1 Cataloging data. For cataloging data purposes, PIN code numbers are assigned as follows:

a. PIN code for type.
   1 - Type I
   2 - Type II

b. PIN code for style.
   A - Style A
   B - Style B

c. PIN code for class.
   1 - Class 1
   2 - Class 2
   3 - Class 3
   4 - Class 4

d. PIN code for application.
   A - Application A
   B - Application B

6.6 Cross reference of classification. The classification of the filter assembly in this document differs from the superseded specification in the following respects:

MIL-F-17690D	MIL-F-17690E
Type I	Type I
Type II	Type II
Not designated	Style A
Not designated	Style B
Class 1	Class 1
Class 2 - with heater	Class 2 - with electric heater
Not designated	Class 3
Not designated	Class 4
MIL-F-17690D	MIL-F-17690E
Not designated	Application A
Not designated	Application B

6.7 Sampling for quality inspection. Recommended inspection level is S-2 with an Acceptable Quality Level (AQL) of 0.05 percent defective (see 4.3).

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6.8 Sampling for packaging inspection. Recommended inspection level is S-2 and an Acceptable Quality Level (AQL) of 2.0 percent defective (see 4.6).

6.9 Subject term (key word) listing.

Filter
Impurities
Lubricating Oil
Turbine Oil
Diesel Engine Oil

6.10 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:

Army - ME
Navy - YD

User Activity:

Coast Guard - CG

Preparing Activity:

Navy - YD

(Project 4330-0058)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

   The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-17690E	2. DOCUMENT DATE (YYMMDD) 29 July 1991

3. DOCUMENT TITLE
MILITARY SPECIFICATION, FILTER ASSEMBLY, FLUID

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
COMMANDING OFFICER	(805) 982-5843	551-5843
c. ADDRESS (Include Zip Code) Naval Construction Battalion Center CESO (Code 1564D/R. Pamplona) Port Hueneme, CA 93043-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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